Commercial processes

National 5 · Materials & manufacture · Worksheet

D&M WITH MR KELLY

Name _____ Class _____ Date _____

By the end of this worksheet I can…

☐ describe the uses and identifying features of injection moulding, die casting, rotational moulding, vacuum forming and sand casting

☐ describe the features of a good vacuum-forming and sand-casting pattern

I DO — Worked example

Read how a full-mark answer is built

A large plastic storage box with a hollow, double-skinned lid.

1 Name the process used to make the hollow lid and outline two reasons it is suitable. (3 marks)

Full-mark answer each ✓ is one mark

Rotational moulding ✓. It produces a hollow, seamless lid in one piece ✓ with an even wall thickness and a good surface finish ✓.

Why marks are not awarded

"Injection moulding because it is good for mass production."

0 marks: injection moulding cannot make a closed hollow form, and "mass production" repeats the question.

WE DO — Guided practice

Use the starters and word bank to help you

A cast iron garden weight.

2 Describe two features that show the weight was sand cast. (2 marks)

Sentence starters

First, …

Next, … using a …

Word bank

fettling · texture · fillets · runner · angles · surface · rounded · corners

Tip: 2 marks = 2 separate, developed points.

YOU DO **On your own** No notes. Check your answers afterwards.

3 Quick check: tick one box for each.

a) A cast-iron manhole cover with a rough texture was most likely made by...
☐ Sand casting ☐ Die casting ☐ Injection moulding ☐ Vacuum forming

b) Why must a vacuum-forming pattern have draft angles?
☐ To make the formed sheet stronger and stiffer ☐ To give the product a smoother surface finish ☐ To make the walls of the product thicker ☐ So it can be removed from the formed sheet

c) Which process suits a toy car body in zinc alloy made in large numbers?
☐ Sand casting ☐ Die casting ☐ Vacuum forming ☐ Rotational moulding

d) Small circular marks on the inside of a plastic casing are evidence of...
☐ Flashing ☐ Sand grains ☐ Ejector pins ☐ Layer lines

4 Key terms: write the letter of the matching description next to each term.

1. **Injection moulding**	____	**A** Large hollow seamless plastic forms: kayaks, bins.
2. **Die casting**	____	**B** High-volume complex metal parts in aluminium/zinc. Same features as injection moulding.
3. **Rotational moulding**	____	**C** Thermoplastic sheet pulled over a pattern: thin walls, draft angles.
4. **Sand casting**	____	**D** High-volume complex thermoplastic parts. Ejector pin marks, injection point, flash.
5. **Vacuum forming**	____	**E** Low-volume metal parts with rough sandy texture.
6. **Vent holes**	____	**F** Allow trapped air out of hollows in a vacuum-forming pattern.

A games controller casing.

5 State two features that show the casing was injection moulded. (2 marks)

A chocolate box insert.

6 State two features that show the insert was vacuum formed. (2 marks)

Zinc alloy toy cars.

7 Outline two reasons die casting is suitable for the toy cars. (2 marks)

Check and reflect

Mark your answers with the answer sheet. Go back to the checklist on the site and re-rate each “I can” statement above. Which one still needs work?